ClipSAT : ACT International Subject Test: Mathematics 2 Level 2

Unit 4.01: Advanced Functions, Polynomials & Logarithms

Name: _____

Q1. The graph of the function fx = x - 1^2 - 4 is shown in the standard x, y coordinate plane below. What is the range of f ?

[ ] A: [-4, infty
[ ] B: [1, infty
[ ] C: -infty, -4]
[ ] D: -infty, infty

Q2. The polynomial function Px = 2x^3 + ax^2 - 5x + b has x - 1 as a factor. When Px is divided by x + 2 , the remainder is -18 . What is the remainder when Px is divided by x - 3 ?

[ ] A: 2
[ ] B: 8
[ ] C: 18
[ ] D: -30

Q3. A function f is defined for all real numbers except x = 0 and x = 1 . For all values of x in its domain, f satisfies the functional equation fx + fleftfrac{x - 1}{x}right = 1 + x. What is the value of f2 ?

[ ] A: frac{3}{4}
[ ] B: frac{5}{4}
[ ] C: frac{7}{4}
[ ] D: frac{9}{4}

Q4. Given the functions fx = 3x - 4 and gx = x^2 + 5 , what is the value of fg2 ?

[ ] A: 23
[ ] B: 9
[ ] C: 18
[ ] D: 27

Q5. The graph of the quadratic function fx = ax^2 + bx + c is shown in the standard x, y coordinate plane below, with vertex 2, -1 and y-intercept 0, 3. The function g is defined by gx = fx + k - 8 , where k is a constant. If the graph of g passes through the origin 0, 0, which of the following could be the value of k ?

[ ] A: -5
[ ] B: -3
[ ] C: 1
[ ] D: 5

Q6. What is the minimum value of the function fx = sqrt{x^2 - 4x + 13} + sqrt{x^2 - 14x + 130} for all real numbers x?

[ ] A: 12
[ ] B: 4sqrt{10}
[ ] C: 13
[ ] D: sqrt{170}

Q7. The function f is defined by fx = frac{3x + 2}{x - 4} for all real numbers x neq 4 . What is the value of f^{-1}2 ?

[ ] A: -10
[ ] B: -6
[ ] C: -4
[ ] D: 6

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Q8. The polynomial function Px = x^3 + ax^2 + bx + 12 , where a and b are real constants, has x - 1 as a factor. When Px is divided by x - 2 , the remainder is -10 . What is the greatest root of the equation Px = 0 ?

[ ] A: 4

[ ] B: 3

[ ] C: 1

[ ] D: 6